

Leon Meng

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EDUCATION

University of California, Irvine

Bachelor of Computer Science and Engineering

Irvine, CA

2025 - 2029

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript, TypeScript, HTML, CSS, SQL

Web & Backend: React, Vite, Tailwind, FastAPI, Supabase, PostgreSQL, Vercel serverless functions, REST APIs

AI & Automation: Anthropic Claude & OpenAI APIs, image processing pipelines, LLM tooling, workflow automation

Data & Quantitative: pandas, NumPy, scikit-learn, SciPy, SQLite, Streamlit, backtesting & walk-forward validation

Robotics & Vision: Odometry & localization, path planning (Roadrunner), finite state machines, OpenCV, Limelight 3A vision, CAD (Fusion 360), photogrammetry (RealityScan), 3D printing

Developer Tools: Git, GitHub, Docker, Linux, CI/CD pipelines, pytest, vitest

EXPERIENCE

SomaFuture

Software Engineer

Oeiras, Portugal

Jun 2026 - Aug 2026

- Designed and built a completely new company website (live demo) that redefined SomaFuture's strategic direction toward standardizing and digitalizing its product line, translating a physical catalog into a structured digital experience
- Built internal mini-apps and automation programs — including file translators that preserve the original document formatting — eliminating manual reformatting and accelerating office workflows
- Prototyped a mock-up 3D room configurator and evaluated competing methods for reconstructing product geometry, comparing photogrammetry (RealityScan) against CAD-authored models to determine a viable digitalization pipeline
- Defined the company's roadmap for incorporating AI into its products and initiated outreach to potential collaborators, including AiHouse, on tooling for floor-plan-to-3D generation and custom home configuration in 3D space

First Tech Challenge — Teams 16031 & 16205

Autonomous & Mechanical Engineer

Vancouver, BC

Oct 2021 - Jun 2025

- Implemented an odometry system using encoders; computed horizontal/vertical displacement per time step, corrected for rotational drift, and converted to field coordinates via trigonometric transforms for sub-centimeter localization accuracy
- Built a vision-guided object detection system using Limelight 3A and OpenCV: designed a color-isolation pipeline (contour detection, color filtering) to detect and localize game elements in real time, then computed target contour angle and position to autonomously align the robot for pickup
- Implemented a finite state machine coordinating all robot mechanisms via event triggers, and integrated Roadrunner for trajectory planning and tuning to produce repeatable multi-step autonomous routines
- Designed and fabricated custom 3D-printed mechanical components (chassis, drivetrain, arm pivot, scoring mechanism) and co-authored the team's Engineering Portfolio; earned multiple top-scoring awards at the British Columbia and Alberta championships

PROJECTS

GARDEROBE

the-garderope.com

Full-Stack Wardrobe Manager — React, Supabase, FastAPI, Claude API

- Built and shipped to production a full-stack wardrobe management app with AI auto-tagging that extracts item name, brand, color, and type from an uploaded photo via the Anthropic Claude API
- Implemented in-browser ML background removal, a drag-and-drop outfit builder, and a live price-tracking wishlist backed by scheduled FastAPI scraping jobs (Grailed, SSENSE) refreshed every 6 hours
- Built a full social layer on Supabase realtime — profiles, publishing, an Explore page, a people directory, friends, and likes — plus a feed aggregating fashion and culture articles from multiple third-party news APIs

Component Style Studio

github.com/ggttlplp201/UI-Library

Visual Page Builder for React Codebases — TypeScript, React, Vite, Tailwind, AST tooling

- Built a local-first visual page builder in which every block is a live React component: scans any React project folder in place, extracts props via react-docgen-typescript, and renders each component in a per-project child dev server so previews use its real CSS, Tailwind config, and dependencies
- Wrote an AST edit engine that writes visual changes (style, text, animation) back into the component's actual source without touching program logic, and exports either a self-contained single-file HTML site or a diffable zip of the edited source
- Designed a multi-page authoring model with a node-graph root connecting pages, per-page loading screens and cursor effects, and a live preview mode; ships a bundled library of 130+ components across three design themes

CS2 Quant — Skin Trading Agents

github.com/ggttlplp201/cs666

Quantitative Research System — Python, pandas, scikit-learn, SQLite, Streamlit

- Built two independent paper-trading agents on the CS2 skin market — one event-driven (trading update-induced repricings), one positional value/trend — over a shared data layer, indicator library, signal bus, and provenance-logged ledger
- Implemented a T+7-aware settlement model, market regime classifier, risk gate, and honest walk-forward backtester that quarantines in-sample results from out-of-sample scoring; covered by a 186-test suite
- Shipped a read-only Streamlit research dashboard surfacing poller data-health gaps, spread-vs-liquidity trading costs, per-rule out-of-sample scorecards, and a browsable decision log tracing why each signal fired

AWARDS & ACHIEVEMENTS

- USACO Gold Division (2023)